Flutter Widget Documentation: CustomStepIndicator

Overview

`CustomStepIndicator` is a reusable Flutter widget that visually displays progress through multiple steps in a process (for example: form steps, booking flow, onboarding, etc.).

It shows:

- A horizontal progress line
- Circular step indicators
- Step labels
- Completed step check icons

This pattern is commonly used in:

- Multi-step forms
- Checkout flows
- Appointment booking
- Onboarding processes

Widget Constructor

```
const CustomStepIndicator({
  super.key,
  required this.currentStep,
  required this.steps,
});
```

Parameters

Parameter	Type	Description
currentStep	int	The currently active step
steps	List<String>	Labels for each step

Example Usage

```
CustomStepIndicator(
  currentStep: 2,
  steps: [
    "Info",
    "Details",
    "Payment",
    "Complete",
  ],
)
```

Result:

- Step 1 → Completed
- Step 2 → Active
- Step 3 → Pending
- Step 4 → Pending

Layout Structure

The widget layout is built using a `Stack`.

```
Stack
 ├── Background line
 ├── Progress line
 └── Steps row (circle + label)
```

This allows the progress line to appear behind the circles.

Step-by-Step Logic

1️⃣ Background Line

```
Positioned(
  top: 16.h,
  left: 20.w,
  right: 20.w,
  child: Container(
```

```
    height: 2,
    color: AppColors.cE5E7EB,
  ),
)
```

Purpose:

Creates the full grey line across all steps.

2️⃣ Progress Line

```
Row(
  children: List.generate(steps.length - 1, (index) {
    return Expanded(
      child: Container(
        height: 2,
        color: index < currentStep - 1
            ? AppColors.c0D6EFD
            : Colors.transparent,
      ),
    );
  }),
)
```

Logic:

```
If step is completed
→ show blue progress line

Else
→ keep transparent
```

Example:

Steps = 4 CurrentStep = 3

Visible progress:

```
Step1 ── Step2 ── Step3 ── Step4
      ✔              ✔       •       ○
```

3️⃣ Generate Steps

```
List.generate(steps.length, (index) {
```

This dynamically creates all steps.

Each step calculates:

```
stepNum = index + 1
```

4️⃣ Step State Detection

```
final isActive = stepNum == currentStep;
final isCompleted = stepNum < currentStep;
```

Possible states:

State	Condition
Completed	stepNum < currentStep
Active	stepNum == currentStep
Pending	stepNum > currentStep

Circle UI

```
Container(
  width: 32.w,
  height: 32.w,
  decoration: BoxDecoration(
    shape: BoxShape.circle,
    color: isActive || isCompleted
        ? AppColors.c0D6EFD
        : const Color(0xFFF3F4F6),
  ),
)
```

Meaning:

Step State	Circle Color
Completed	Blue
Active	Blue
Pending	Light grey

Completed Icon

```
isCompleted
  ? Icon(Icons.check_rounded)
```

If the step is completed:

- A check icon appears

Otherwise:

- The step number is displayed

Step Number

```
Text('$stepNum')
```

Displayed only when:

```
step not completed
```

Step Label

```
Text(
  steps[index],
)
```

This shows the label below the step circle.

Example labels:

```
Info
Details
Payment
Complete
```

Color Logic for Labels

```
color: stepNum <= currentStep
    ? AppColors.c0D6EFD
    : AppColors.c9CA3AF
```

Meaning:

Step	Label Color
Completed	Blue
Active	Blue
Pending	Grey

Visual Example

```
✔️   ✔️  ●  ○
Info Details Payment Complete
```

Legend:

```
✔️ Completed
● Active
○ Pending
```

Advantages

- Fully reusable
- Dynamic step count
- Supports any label
- Clean UI
- Works with responsive sizing

Possible Improvements

You can extend this widget with:

- Tapable steps
- Animated progress line
- Step icons instead of numbers
- Vertical step indicator
- Gradient progress line

Full Source Code

```dart
import 'package:thedoctor0101/constants/common_imports.dart';

class CustomStepIndicator extends StatelessWidget {
  final int currentStep;
  final List<String> steps;

  const CustomStepIndicator({
    super.key,
    required this.currentStep,
    required this.steps,
  });

  @override
  Widget build(BuildContext context) {
    return SizedBox(
      height: 80.h,
      child: Stack(
        children: [
          Positioned(
            top: 16.h,
            left: 20.w,
```

```
      right: 20.w,
      child: Container(
        height: 2,
        color: AppColors.cE5E7EB,
      ),
    ),

    Positioned(
      top: 16.h,
      left: 20.w,
      right: 20.w,
      child: Row(
        children: List.generate(steps.length - 1, (index) {
          return Expanded(
            child: Container(
              height: 2,
              color: index < currentStep - 1
                  ? AppColors.c0D6EFD
                  : Colors.transparent,
            ),
          );
        }),
      ),
    ),

    Row(
      mainAxisAlignment: MainAxisAlignment.spaceBetween,
      children: List.generate(steps.length, (index) {
        final stepNum = index + 1;
        final isActive = stepNum == currentStep;
        final isCompleted = stepNum < currentStep;

        return Column(
          mainAxisSize: MainAxisSize.min,
          children: [
            Container(
              width: 32.w,
              height: 32.w,
              decoration: BoxDecoration(
                shape: BoxShape.circle,
                color: isActive || isCompleted
                    ? AppColors.c0D6EFD
                    : const Color(0xFFF3F4F6),
              ),
              child: Center(
                child: isCompleted
                    ? Icon(
                        Icons.check_rounded,
```

```
                            size: 14.sp,
                            color: Colors.white,
                          )
                        : Text(
                            '$stepNum',
                            style: TextFontStyle
                                .textStyle14w400c1F2937SFPro
                                .copyWith(
                              fontSize: 13.sp,
                              fontWeight: FontWeight.w600,
                              color: isActive
                                  ? Colors.white
                                  : AppColors.c9CA3AF,
                            ),
                          ),
                    ),
                  ),

                  SizedBox(height: 5),

                  SizedBox(
                    child: Center(
                      child: Text(
                        steps[index],
                        textAlign: TextAlign.center,
                        style: TextFontStyle
                            .textStyle14w400c1F2937SFPro
                            .copyWith(
                          fontSize: 11.sp,
                          fontWeight: isActive
                              ? FontWeight.w600
                              : FontWeight.w400,
                          color: stepNum <= currentStep
                              ? AppColors.c0D6EFD
                              : AppColors.c9CA3AF,
                        ),
                      ),
                    ),
                  ),
                ],
              );
            }),
          ),
        ],
      ),
    );
  }
}
```